

Module 51 – Promises

1. What is a Promise?

A Promise is a JavaScript object used to handle an asynchronous operation and represent its future result, which can be either successful or failed.

Promise States:

- **Pending** → Operation is still in progress.
- **Fulfilled** → Operation completed successfully.
- **Rejected** → Operation failed.

Promise Methods:

- **.then()** → Handles a successful result.
- **.catch()** → Handles an error or rejected result.
- **.finally()** → Executes after the Promise is completed, whether successful or failed.

Example Code:

```
let newPromise = new Promise((resolve, reject) => {  
  let dataReceived = true;  
  if(dataReceived){  
    resolve("Data Received");  
  }  
  else{  
    reject("Data Not Received");  
  }  
});  
  
newPromise  
  .then((message) => {  
    console.log("Success: " + message);  
    return "Next Success " + message;  
  })  
  .then((nextMessage) => {  
    console.log(nextMessage);  
  })  
  .catch((error) => {  
    console.log("Failure: " + error);  
  })  
  .finally(() => {  
    console.log("End");  
  });
```